

Panel D densitometry — your method, run reliably in Fiji

The reading you described is the correct and only reliable one: for each lane, measure the over-expressed product band in the 15–20 kDa window, and normalise it to a steady high-molecular-weight band at the top of the *same* lane, so each lane is measured against itself. This cancels loading and total-protein differences (which is why raw band intensity is meaningless — more host protein at 37 °C inflates it).

Software could not place the 24 uneven lanes on these scans accurately (it kept finding ~20 and the product box drifted onto the wrong band). The fix is to place the lanes by hand once, in Fiji, which takes a couple of minutes per gel and makes the measurement trustworthy. The four gels are already MW-calibrated with your 20/15 kDa marks (gel1-4_calibrated_*.png).

One-time setup

Install Fiji (ImageJ) if not already on the machine (free, imagej.net/software/fiji/). Open the gel scan (the _slide_gels/image40–43 files, or the originals in secretion/).

Per gel — the steps

1. Convert to 8-bit greyscale: Image > Type > 8-bit. Then Edit > Invert is not needed — Fiji's gel tool handles dark bands on light background.
2. Draw lane 1: select the rectangle tool, draw a tall narrow box covering the full height of the first lane (from above the top steady band down through the 15 kDa line). Analyze > Gels > Select First Lane (or press 1).
3. Move the same box to lane 2, Analyze > Gels > Select Next Lane (press 2). Repeat for all 24 lanes, left to right, following the lane key (blocks of 8: 16 / 22 / 37 °C).
4. Analyze > Gels > Plot Lanes. Fiji draws an intensity profile per lane.
5. In each lane's profile you will see two peaks that matter: the steady top band (your reference) and the product band in the 15–20 kDa window. Use the line tool to close the base of each peak, then the wand tool to measure each peak's area.
6. For each lane compute: product peak area ÷ top-band peak area. That ratio is your self-normalised expression readout.

Turning ratios into the Panel D grid

Average the ratio across the four peptides within each signal-temperature cell, then bin into your four-level scale (no expression / weak / medium / strong) by eye or by simple thresholds on the ratio. That fills the grid honestly, with your own normalisation.

Gel-to-condition key (recap)

- Gel 1 — DsbA + MalE, 16 / 22 / 37 °C. Lanes per block: D1 M1 D2 M2 D3 M3 D4 M4.
- Gel 2 — OmpA + PhoA, 18 / 22 / 37 °C. O1 P1 O2 P2 O3 P3 O4 P4. (Confirm the 37 °C block — slide label looks mis-pasted.)
- Gel 3 — PelB + VNP15, 16 / 22 / 37 °C. P1 V1 P2 V2 P3 V3 P4 V4.
- Gel 4 — NS control, 16 / 22 / 37 °C. NS1–NS4 per block. Product ~14–15 kDa (runs lower — no signal peptide).

What the rough automated pass suggested (not certified)

Even with imperfect lane placement, Gel 1 showed the 37 °C block collapsing (several lanes with no product band), against stronger 16 and 22 °C blocks — consistent with your recollection that 37 °C gives lower yield and more degradation. Treat this as a hint of the expected direction, not a result; the Fiji pass above is what produces the real numbers.

The honest alternative

Panel D does not have to be quantitative. Because the paper's real quantification is A280/SEC (Figure 4), Panel D can simply present the best-condition call, scored by eye from the calibrated gels. That is a legitimate methods-paper choice and avoids over-claiming precision the gels cannot support.